

Geometry Basics

AI Tutor - Mathematics - Lesson 3

Basic Shapes

Geometry studies shapes, their properties, and the space they occupy. Every shape is built from fundamental elements: points, lines, and angles.

- **Triangle** - 3 sides, angles sum to 180 degrees.
- **Square** - 4 equal sides, 4 right angles (90 degrees each).
- **Circle** - all points equidistant from a center point.

Angles

An angle is formed where two lines meet. Angles are measured in degrees.

- **Acute angle** - less than 90 degrees.
- **Right angle** - exactly 90 degrees.
- **Obtuse angle** - more than 90 but less than 180 degrees.

Perimeter and Area

Perimeter is the total distance around a shape's boundary. Area is the amount of space inside the shape.

- Rectangle perimeter = $2 \times (\text{length} + \text{width})$
- Rectangle area = $\text{length} \times \text{width}$
- Circle area = $\pi \times \text{radius}^2$